



examGenome CheatSheet on Algebra

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1	What is a ratio? Answer: A ratio compares two or more quantities by showing how much of one thing there is compared to another. Explanation: For example, if there are 3 red marbles and 2 blue marbles, the ratio of red to blue is 3:2. Ratios use colons (:) or the word 'to'.
2	Do ratios have units? Answer: No—ratios compare quantities, so the units cancel out as long as both quantities use the same unit. Explanation: For example, 10 cm : 5 cm becomes 10:5, which simplifies to 2:1. The 'cm' is not written in the final ratio.
3	How do you simplify a ratio? Answer: Divide both numbers by their greatest common factor (GCF), just like simplifying fractions. Explanation: For example, 8:12 → divide both by 4 → 2:3. The simplest form uses the smallest whole numbers possible.
4	What does the order in a ratio mean? Answer: Order matters! 'Red to blue = 3:2' is not the same as 'blue to red = 2:3'. Explanation: The first number always matches the first item mentioned. Always read or write ratios in the correct order.
5	Can a ratio compare more than two quantities? Answer: Yes! For example, a fruit salad with 2 apples, 3 bananas, and 1 orange has a ratio of 2:3:1. Explanation: You can have three or more parts in a ratio—as long as you keep the order clear.
6	What is the difference between a ratio and a fraction? Answer: A fraction compares a part to the whole (e.g., $\frac{3}{5}$ of the class are girls), while a ratio compares part to part (e.g., girls to boys = 3:2). Explanation: Ratios can also be written as fractions, but their meaning depends on context—part-to-part vs. part-to-whole.
7	Can ratios be written in different forms? Answer: Yes! Ratios can be written as 3:2, 3 to 2, or even as a fraction $\frac{3}{2}$ (when comparing part to part). Explanation: All three mean the same comparison—just different ways to write it.
8	What happens to a ratio if you multiply or divide both parts by the same number? Answer: The ratio stays the same—it's equivalent, just not in simplest form. Explanation: 2:3 is the same as 4:6 or 10:15 because both parts were multiplied by the same number.

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9	Can a ratio include zero? Answer: Only if it makes sense—like 5:0 means '5 of something and none of the other.' But 0:0 is not allowed. Explanation: For example, if a team scored 5 goals and the opponent scored 0, the score ratio is 5:0. But you can't simplify a ratio that has zero in it.
10	When are two ratios equivalent? Answer: When they simplify to the same simplest form. Explanation: For example, 6:9 and 4:6 are both equivalent to 2:3, so they are equal ratios.

